

Systematic futures: method comparison under one evaluation protocol

Research note

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Abstract

Sixteen CME futures roots, daily contract-level prices from 2010 to 2024Q1, six classic method families and two LightGBM variants, one vectorised engine, one volatility overlay, one cost model and one evaluation protocol: a pre-declared decision rule inside develop+validate, and a single out-of-sample window read once. The headline result is negative in the useful direction: the ML variants that win the pre-declared comparison in-sample fail it out of time, while turnover and the diversification return decide the rest. Every number below is read from the committed result files in `results/`; market data is not redistributed.

1 Data and construction

Raw per-contract daily prices for 16 CME roots from pssystemtrade’s `multiple_prices_csv` snapshot, pinned at commit `b4a25e6` and frozen upstream on 2024-03-28. Each row carries the three active legs (front, successor, prior) with contract identifiers. The roll calendar is *observed* from the leg data rather than estimated; the continuous series is built by ratio back-adjustment (at each roll, the new-to-old front ratio on the prior session rescales earlier closes), with the invariant that the continuous series’ daily return equals the held contract’s own return on every session including roll days, pinned by test at floating-point tolerance. The basis series is computed from raw closes as $\text{close}(\text{next})/\text{close}(\text{front}) - 1$. Every derived file is SHA-256-gated by a committed manifest; ingest fails on drift.

2 Method specifications

Family	Rule	Source
Baseline	constant full-long, overlay-sized (null model)	M4
TSMOM	equal-weight vol-scaled 5/10/20d + 1/3/12m horizon signs	M1, M2, S4
XS momentum	centred cross-sectional rank of 12-month return	S3
Carry	sign of the negative basis (long backwardated)	M3
Seasonality	sign of same-calendar-month trailing score; and a $\pm 50\%$ tilt on TSMOM	S4
6a <code>ml_defaults</code>	LightGBM defaults, purged/embargoed walk-forward, no search	M5–M7
6b <code>ml_tuned</code>	as 6a with a pruned TPE search per fold	M5–M7

3 Evaluation protocol

One accounting path (`engine.run`) for every family and every parameter search: signals in $[-1, 1]$ per symbol per day, a shared sizing overlay that scales *each symbol* to a 10% annualized volatility target with ± 1 notional caps applied identically, a linear cost charge of b basis points per side on traded notional, and metrics reported at $b \in \{0, 2, 5, 10\}$ with $b = 2$ the headline level a decision

is read at. Turnover is book-level: mean daily traded notional per unit of capital, the base the charge is taken on, so b costs $\text{turnover} \times b \times 252 \times 10^{-4}$ of annual return. Sharpe is annualized at $\sqrt{252}$ in both windows as a stated convention: the frozen panel carries 309.6 sessions per year before 2022 and 258.4 after, so the two windows' Sharpe *levels* are not strictly comparable, while the DSR the rule reads is a probability and is unaffected. Sizing per symbol is not sizing per book: realized book volatility runs 0.9–3.0% in develop+validate and 1.5–4.6% out of time, so the families are like-for-like in rule and cost, not in delivered risk. A signal formed with data through day t drives the position held from t to $t + 1$; a regression test fails on a look-ahead implementation and passes on the correct convention. Develop 2010–2019 and validate 2020–2021 form the decision window; the out-of-sample window 2022–2024Q1 is read exactly once, at the end, by one script, with the walk-forward schedule extended across the whole panel so that the fold covering 2022 trains entirely inside develop+validate. Reports carry Sharpe, Sortino, maximum drawdown, turnover and the Deflated Sharpe Ratio (deflated by the configurations each selection actually evaluated: 1 for every classic family, 100 for the tuned variant).

4 Results

Family	Develop+validate		Out-of-sample	
	Sharpe	DSR	Sharpe	DSR
Baseline	0.408	0.9403	−0.694	0.1470
TSMOM (benchmark)	−0.085	0.3726	0.551	0.7966
XS momentum	0.406	0.9397	1.174	0.9611
Carry	−3.614	0.0000	−1.756	0.0041
Seasonality	−0.088	0.3683	0.770	0.8793
Seasonal tilt	−0.116	0.3284	0.407	0.7304
6a ml_defaults	0.799	0.9989	0.072	0.5433
6b ml_tuned	1.550	0.9999	0.390	0.0261

Table 1: Sharpe and Deflated Sharpe Ratio at 2 bps/side. Pre-declared rule: an ML variant wins only if its decision-window DSR beats the benchmark's. Both clear it in develop+validate; both fail it out of time.

Family	0 bps	2 bps	5 bps	10 bps
Baseline	0.419	0.408	0.390	0.361
TSMOM	0.200	−0.085	−0.508	−1.203
XS momentum	0.562	0.406	0.172	−0.216
Carry	−3.214	−3.614	−4.203	−5.147
Seasonality	−0.027	−0.088	−0.177	−0.326
Seasonal tilt	0.163	−0.116	−0.530	−1.208
6a ml_defaults	1.406	0.799	−0.122	−1.660
6b ml_tuned	2.074	1.550	0.746	−0.616

Table 2: Develop+validate Sharpe by cost level. Turnover is identical across a row; only the price per trade changes. Trend, the tilt and both ML variants flip sign between gross and headline cost.

5 Findings

1. **Costs select strategies.** At 5 bps the survivors differ by window: develop+validate leaves the baseline, XS momentum and the tuned variant positive, while the out-of-sample window

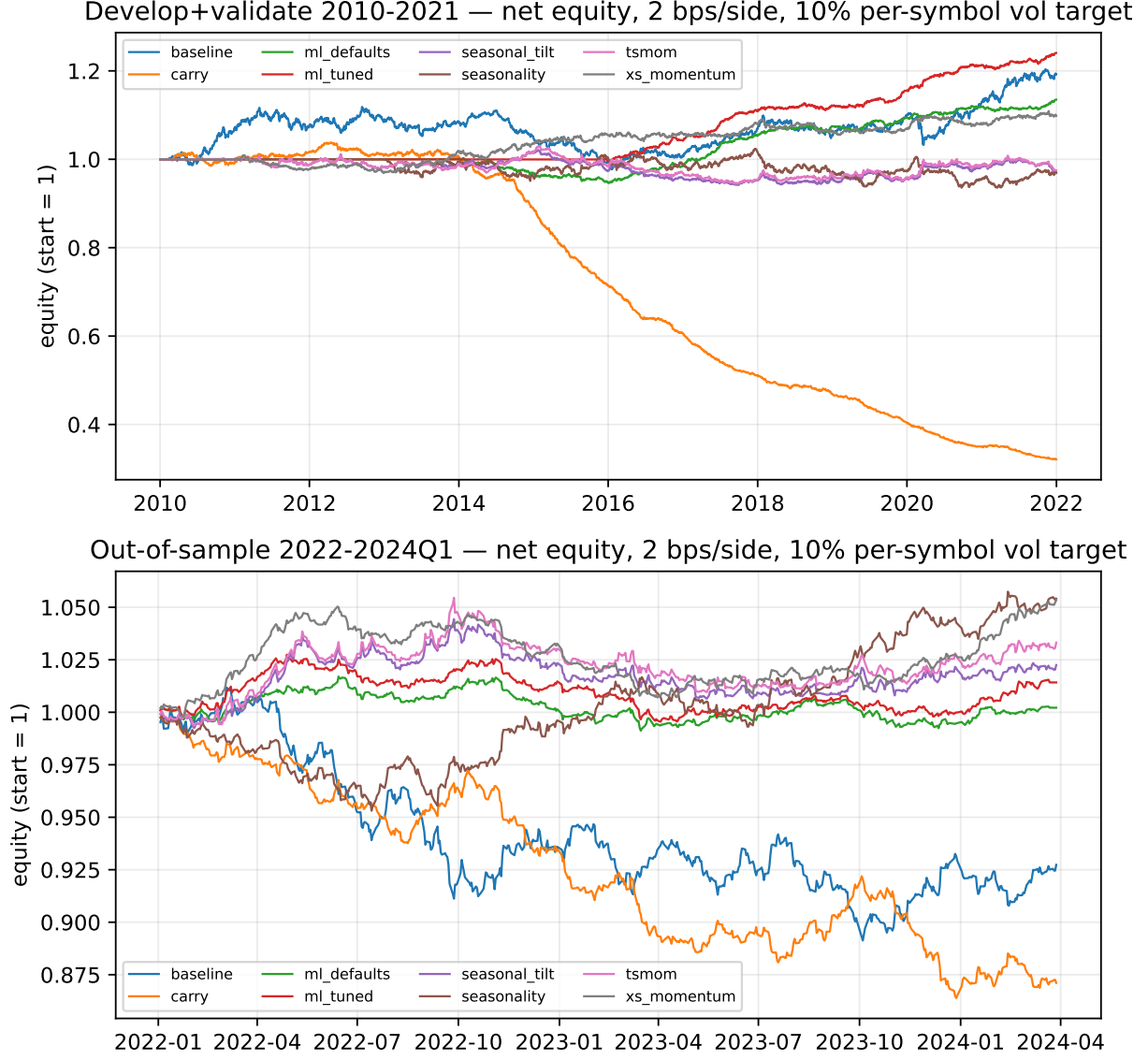


Figure 1: Net equity at 2 bps/side, both windows, from the committed return series.

leaves XS momentum, seasonality, the trend benchmark and the tilt. Trend and the tilt flip sign between gross and the headline level; both ML variants stay positive on both, at a cost of most of their signal.

2. **A null model is a real competitor.** Vol-targeted buy-and-hold earns 0.408 at 2 bps on book turnover of 0.007, which is the diversification return any signal must beat before claiming edge.
3. **The ML result is the honest one.** In-window winners (DSR 0.9989 / 0.9999 against the benchmark's 0.3726), out-of-time failures (0.5433 / 0.0261 against 0.7966). The in-window advantage is concentrated in 2015–2019 and 2020–2021, the periods the benchmark lost money, while 6a was -0.80 in 2010–2014 and 6b had no coverage there at all. Out of time, cost consumes 6a's signal: 0.759 gross, 0.072 net, book turnover 0.199 per day.
4. **Purge and embargo guarantee no label leakage, not generalisation.** The folds are clean by construction and asserted on every run; the features are a non-linear re-expression of the same premia, so the protocol could not create persistence the premia do not have. A

naive-split counterfactual was deliberately not run: the size of what the protocol prevented is argued, not measured.

5. **Standalone seasonality fails on signal, not cost** (-0.027 gross at book turnover 0.025 , a quarter of the benchmark's 0.103), on a score backed by five priors per calendar month.
6. **Carry's number is a frequency artefact.** The published construction is a slope magnitude with rank weighting and monthly rebalancing; what ran here is the sign of the current basis, refreshed daily, and that sign moves with the front leg's own price. Holding the identical signal from each month's first session earns $+0.163$ where the daily refresh earns -3.614 , and inverting the daily rule earns $+2.807$ (`results/out_of_sample/carry_frequency.csv`). The sign convention is *not* resolved by adopting the profitable direction, and the carry premium is neither confirmed nor refuted by these numbers.
7. **2023 is the year nobody earned.** Out of time, every family sits at or below zero in 2023 except seasonality, while 2022 and 2024Q1 pay trend and XS momentum. The passive book loses in all three, so what generalised was cross-sectional selection, not exposure.

6 Limitations

One universe (16 CME roots), one out-of-sample window, one read. The cost model is linear and constant: no market impact, no bid/ask, no volume-dependent slippage. M6's CSCV probability-of-backtest-overfitting diagnostic is not implemented; the multiple-testing discipline is the trial-count DSR deflation plus the single out-of-sample read. Two SHOULD readings (S5, S9) and the M7 book chapters are not read: the interpretation resting on them is marked as scaffolded in `docs/findings/memo.md`. The ML result is specific to this feature set, label horizon, fold geometry, cost level and window; it is not a statement about machine learning on futures in general.

Reproducibility

`uv sync`, fetch the pinned snapshot, rebuild derived data (manifest-gated), `uv run python scripts/build_family/build_decision.py, build_out_of_sample.py`; the committed tables are reproduced exactly. Reading order for the story: `notebooks/analysis.ipynb`, then `docs/findings/memo.md`.